

Basic Mechanical Engineering (ME2001)

Date: June 1st 2024

Course Instructor(s)

1. Dr. Kashif Saeed
2. Mohsin Yousuf (CM)

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 5

Q1	Q2	Q3	Q4	Q5
10	10	5	10	15

Roll No. _____ Section _____

Student Signature _____

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1. Attempt all the questions, programmable calculators not allowed.
2. Attempt all parts of the same question together.
3. Show all the steps with the help of diagrams and answers with proper units.
4. In questions, hints are mentioned in *italic*.

CLO # 01: Calculate the moment of a force/couple.

Q1: A bent bar is shown in Fig. 1. Determine the resultant moment at A by scalar formulation and verify by using vector formulation. [10 marks]

Remember to mention the support reaction at A in your FBD.

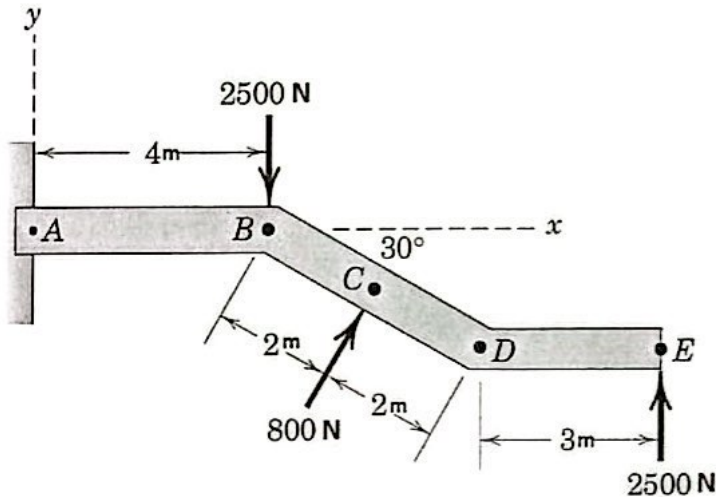


Figure 1. A Bent Bar

CLO # 02: Analyze static equilibrium analysis of a rigid body by applying Newton Laws of Motion and concept of dry friction.

- Q2: The man exerts a force P on the car at an angle $\alpha = 20^\circ$. The 1760-kg car has front wheel drive (FWD). The driver spins the front wheels, and the coefficient of kinetic friction is $\mu_k = 0.02$. Snow behind the rear tires exerts a horizontal resisting force S . Getting the car to move requires overcoming a resisting force $S = 420$ N. Figure out the force P , the man must exert. [10 marks]

The center of gravity point is at 1.62 m from the rear tire as shown in Fig. 2. Remember to draw its FBD first.

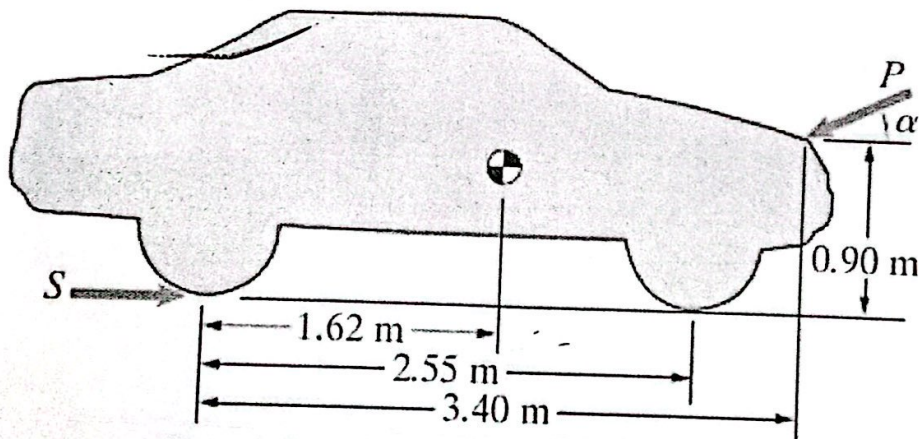


Figure 2. Car in Snow with Centre of Gravity point

CLO # 03: Analyze absolute motion of rigid bodies in general plane motion.

Q3: The slender bar AB in the Fig. 3 shown below is tied to a drum which is rotating with angular velocity ω_0 . **Figure out** the angular velocity ω_{AB} of the slender bar as a function of the distance x and ω_0 .

[5 marks]

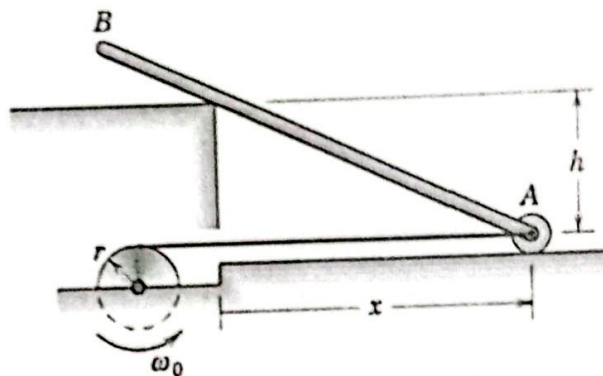


Figure 3. A Slender with Rotating Drum

CLO # 04: Analyze relative motion of rigid bodies in rotation and/or translation.

Q4: Two rotor blades of 800-mm radius rotate counterclockwise about the shaft at O mounted in the sliding block as shown in Fig. 4. The blade with point A completes 200 revolutions in one minute and has an angular acceleration of 3 rad/s^2 . The block moves toward right-hand side with a velocity and acceleration of 4 m/s and 5 m/s^2 , respectively. **Figure out** the magnitude of the velocity and acceleration of the tip A of the blade when $\theta = 30^\circ$.

[10 marks]

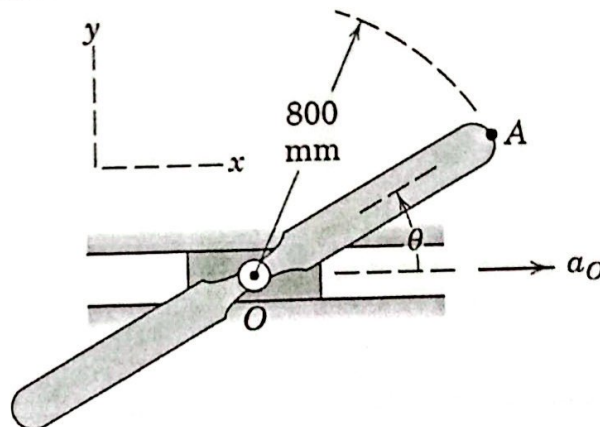


Figure 4. Rotating Blades

CLO # 05: Evaluate the internal forces in the members of a loaded truss and assess safety of the structure.

Q5: A plane truss is shown in the Fig. 5 below. Use the method of joints to **evaluate** the force supported by each member of the truss. **Verify** the force in member BD using method of sections. **[15 marks]**

There is pin at A and rocker support at E.

All members have a square cross-section having a perimeter of 20 cm. The ultimate tensile/compressive stress of the material used is 2 N/mm^2 .

Comment on the safety of the member BD.

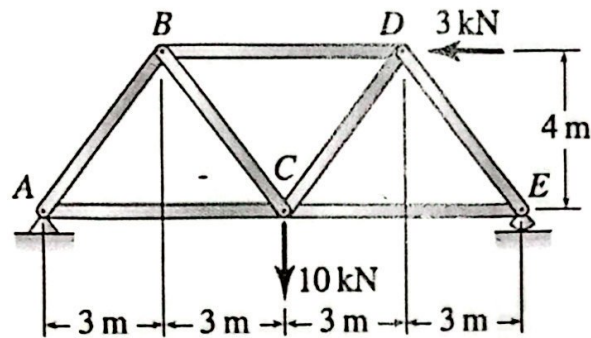


Figure 5. A Plane Truss